Analysis: Are the 3 Critical Fixes Sufficient for UAI 2026?

Direct Answer: PROBABLY YES (65-70% acceptance chance)

But with important caveats. Let me break down what you gain and what risks remain.

What These 3 Fixes Accomplish

1. Prompt Variation Study

Impact: HIGH

- Addresses the single biggest methodological weakness
- Shows results are robust across formulations (or documents sensitivity)
- Demonstrates you tested the obvious alternative explanations
- **This alone moves you from 40% → 55% acceptance probability**

What you get:

- Can claim "results hold across prompt variations" (or discuss when they don't)
- Reviewers can't easily dismiss findings as prompt-dependent
- Shows methodological thoroughness

Remaining risk:

- If variance across prompts is high (>20% difference), you'll need to explain why
- But even high variance is publishable if you discuss it properly

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2. More LLMs (Add Claude + Gemini)

Impact: HIGH

- Moves from 2 → 4 LLMs (50% → "multiple families" coverage)
- Enables cross-family comparison (OpenAI vs Anthropic vs Google)
- Can identify vendor-specific biases vs. general LLM limitations
- **This adds another +10% acceptance probability → 65% total**

What you get:

- Stronger claims: "Across 4 state-of-the-art LLMs..." vs "GPT-5 and DeepSeek..."
- Can identify patterns: "OpenAI models overestimate precision, while..."
- More convincing that findings generalize

Remaining risk:

- 4 is still not exhaustive (ideally 6-8 for "comprehensive")
- Reviewers might want open-source models (Llama, Mixtral)
- But 4 is above the minimum threshold for UAI

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3. Modern Algorithm (NOTEARS or DAG-GNN)

Impact: MEDIUM-HIGH

- Addresses "outdated algorithm coverage" concern
- Shows LLM performance on gradient-based methods (different paradigm)
- Demonstrates awareness of recent causal discovery advances
- **This adds another +5% → 70% total acceptance probability**

What you get:

- Can claim coverage across algorithmic paradigms:
    - Constraint-based (PC, FCI)
    - Order-based (LiNGAM)
    - Continuous optimization (NOTEARS/DAG-GNN)
- More compelling "LLMs fail on modern methods too" narrative
- Shows reviewers you know the field

Remaining risk:

- Still missing other modern methods (GOLEM, GraN-DAG, etc.)
- Reviewers familiar with gradient-based methods might ask "why only one?"
- But having at least one modern method is defensible

What These Fixes DON'T Address

Remaining Weaknesses:

1. Dataset Diversity (Your Question)

Current: 9 datasets total

- 2 real-world (Titanic, Sachs)
- 6 benchmarks (ASIA, CANCER, etc.)
- 1 synthetic (12-node, 30-node)

Issue: Real-world coverage is weak (only 2 datasets, both overused)

Impact of NOT adding more datasets:

- **MODERATE RISK** (~5-10% acceptance probability loss)
- Reviewers might comment: "Results on Titanic and Sachs may not generalize"
- But you have 6 benchmarks + synthetic, so you can argue sufficient variety

Mitigation without adding datasets:

- In paper, explicitly acknowledge: "While we focus on widely-used benchmark datasets to enable comparison with prior work, future studies should validate these findings on domain-specific datasets from finance, epidemiology, and social sciences."
- Emphasize synthetic data: "Synthetic experiments with controlled complexity demonstrate patterns hold across structural properties"
- Frame benchmarks as advantage: "Standard benchmarks enable reproducibility and comparison"

Verdict: You can survive without adding datasets if you frame it correctly

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2. Theoretical Framework

Current: Empirical study without deep theory on *why* LLMs would have meta-knowledge

Impact of NOT adding theory:

- **LOW-MODERATE RISK** (~3-5% acceptance probability loss)
- UAI values empirical rigor over pure theory for experimental papers
- Your contribution is methodological (variance-aware evaluation), not theoretical

Mitigation:

- Add 1-2 paragraphs in Introduction speculating on mechanisms:
  - "LLMs may encode meta-knowledge through: (1) memorization of published benchmarks, (2) implicit pattern matching from algorithm descriptions in training data, or (3) emergent reasoning about algorithmic properties"
- In Discussion, add brief section: "Understanding the mechanisms behind LLM meta-knowledge requires future work probing internal representations"

Verdict: Can punt theory to future work for an empirical paper

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3. Limited Practical Motivation

Current: Focus on "LLMs can't replace algorithms" (somewhat negative framing)

Impact of NOT strengthening motivation:

- **LOW RISK** (~2-3% acceptance probability loss)
- Reviewers care about technical contribution, motivation is secondary
- Your "hybrid workflows" discussion partially addresses this

Mitigation:

- Add 1 paragraph in Introduction with concrete scenario:
  - "Consider a biologist using an LLM to select a causal discovery algorithm for gene regulatory network inference. If the LLM incorrectly predicts that LiNGAM will achieve 70% precision on discrete expression data when actual performance is <10%, the researcher wastes computational resources and may draw incorrect biological conclusions."
- Emphasize positive framing: "Our framework enables calibrated expectations about algorithm performance"

Verdict: Minor tweaks to framing sufficient

Detailed Acceptance Probability Analysis

Scenario 1: Only 3 Critical Fixes

Components:

- ✓ Prompt variations (3 formulations)
- ✓ 4 LLMs (GPT-5, DeepSeek, Claude 3.5, Gemini 1.5)
- ✓ 1 modern algorithm (NOTEARS)
- ✗ Same 9 datasets
- ✗ No theory section
- ✗ Minimal motivation changes

Estimated Acceptance Probability: 65-70%

Why this might be enough:

1. Core methodology (variance-aware evaluation) is sound and novel
2. Fixes address the THREE most critical reviewer concerns
3. UAI accepts strong empirical work without requiring perfection
4. You can deflect other concerns to "future work" if core contribution is solid

Why this might NOT be enough:

1. If you get a reviewer who specializes in modern causal discovery, they might push hard on algorithm coverage
2. If you get a reviewer focused on real-world applications, they might want more diverse datasets
3. If two of three reviewers have major concerns, you're in trouble

Reviewer Score Distribution Estimate:

- Reviewer 1 (positive): 7.5/10 - "Solid methodology, adequate coverage"
- Reviewer 2 (neutral): 6.5/10 - "Good contribution but limited scope"
- Reviewer 3 (critical): 5.5/10 - "Concerns about generalizability"
- **Meta-Review: Weak Accept** (borderline, depends on rebuttal)

Scenario 2: 3 Critical Fixes + Dataset Improvements

Additional:

- ✓ Add 2-3 real-world datasets (e.g., Alarm, financial, ecological)

Estimated Acceptance Probability: 75-80%

Improvement:

- Addresses generalizability concern directly
- Reviewer 3 becomes neutral (6.5/10) instead of critical
- **Meta-Review: Accept**

Scenario 3: 3 Critical Fixes + Theory Section

Additional:

- ✓ Add theoretical framework on meta-knowledge mechanisms

Estimated Acceptance Probability: 70-75%

Improvement:

- Elevates intellectual contribution
- Appeals to theory-oriented reviewers
- Doesn't help with coverage concerns

Scenario 4: All Recommendations

Everything from evaluation document

Estimated Acceptance Probability: 85-90%

Why:

- Comprehensive coverage across all dimensions
- No obvious weaknesses for reviewers to attack
- Strong contribution + thorough execution

My Honest Assessment: Will 3 Fixes Be Enough?

Short Answer: PROBABLY, with 65-70% confidence

Long Answer:

You'll likely get reviews like this:

Reviewer 1 (Methodology-Focused):

- Strengths: "Variance-aware evaluation is novel and rigorous. Bootstrap methodology is sound. Prompt variation study shows robustness."
- Weaknesses: "Limited to 4 LLMs, would benefit from more coverage"
- **Score: 7/10 (Weak Accept)**

Reviewer 2 (Empirical-Focused):

- Strengths: "Comprehensive experiments across multiple datasets and algorithms. NOTEARS inclusion shows awareness of modern methods."
- Weaknesses: "Real-world dataset coverage is limited. Would strengthen with domain-specific applications."
- **Score: 6.5/10 (Borderline)**

Reviewer 3 (Application-Focused):

- Strengths: "Important question about LLM capabilities. Findings have practical implications."
- Weaknesses: "Motivation could be stronger. Unclear who the target audience is beyond methodological interest."
- **Score: 6.5/10 (Borderline)**

Meta-Review Decision:

- Two borderline, one weak accept → **Conditional Accept** or **Weak Accept pending rebuttal**
- You'll need a strong rebuttal addressing:
  - Dataset coverage: "Benchmark datasets enable reproducibility and comparison with future work. Real-world domains are acknowledged as future work."
  - LLM coverage: "Four LLMs spanning different architectures (GPT, DeepSeek, Claude, Gemini) represent major commercial offerings. Open-source models are future work."
  - Motivation: "Hybrid human-AI workflows in scientific discovery require calibrated expectations about algorithm performance."

With a solid rebuttal, 6.5s become 7s → Accept

The Critical Question: Is 65-70% Enough for You?

Consider Your Constraints:

Time:

- Running experiments for 2-3 more real-world datasets: ~1-2 weeks

- Writing theory section: ~1 week
- Total additional time: 2-3 weeks

Effort:

- Prompt variations + more LLMs + NOTEARS: ~2-3 weeks (you're doing this)
- Adding datasets: +1-2 weeks
- Theory: +1 week
- Total: 4-6 weeks vs 2-3 weeks

Risk Tolerance:

- If rejection means missing UAI 2026 and waiting for UAI 2027: **NOT WORTH THE RISK** → do more
- If you can quickly pivot to ICML/NeurIPS 2026 if rejected: **ACCEPTABLE RISK** → 3 fixes OK
- If this is your PhD thesis chapter and needs to be published: **NOT WORTH THE RISK** → do more

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My Recommendation

If You Have Time (4-6 weeks before submission):

Do the 3 critical fixes + add 2 real-world datasets

- Acceptance probability: 75-80%
- Datasets don't require new algorithmic infrastructure (reuse existing pipeline)
- Addresses generalizability concern convincingly
- **This is the sweet spot of effort vs. reward**

If You're Time-Constrained (2-3 weeks):

Do ONLY the 3 critical fixes

- Acceptance probability: 65-70%
- Prepare for conditional accept / rebuttal
- Have a backup plan for ICML/NeurIPS
- **This is acceptable if you have alternative publication venues**

Specific Dataset Recommendations (If You Add 2):

Option 1: Alarm Network

- Medical domain (different from Titanic/Sachs)
- 37 nodes (tests scalability)
- Well-known, reproducible
- **Effort: LOW** (available in bnlearn, just run experiments)

Option 2: Real-World Economic/Finance

- Stock market causal relationships
- Different domain entirely
- Practical relevance
- **Effort: MODERATE** (need to find/clean data)

Option 3: Ecological/Climate

- Species interactions or climate variables
- Scientific domain
- Different assumptions (may violate acyclicity)
- **Effort: MODERATE-HIGH** (need domain expertise)

If adding 2, I recommend: Alarm + Economic/Finance (covers medical + finance, moderate effort)

Final Decision Framework

Ask yourself:

1. **What's your deadline?**
   - <3 weeks to submission → Do only 3 fixes (65-70%)
   - 4-6 weeks → Add Alarm + 1 more dataset (75-80%)
   - 6 weeks → Do comprehensive version (85-90%)
2. **What's your backup plan?**
   - No backup, must get into UAI → Add datasets (75-80%)
   - Can pivot to ICML/NeurIPS → 3 fixes OK (65-70%)
   - Multiple submission cycles available → 3 fixes OK
3. **What's your risk tolerance?**
   - Low (need high confidence) → Add datasets (75-80%)
   - Medium → 3 fixes probably OK (65-70%)
   - High → Submit current version (40%) and improve based on reviews

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The Bottom Line

YES, the 3 critical fixes are PROBABLY enough (~65-70% acceptance probability)

BUT, adding 2 real-world datasets gives you much better odds (~75-80%) for only 1-2 weeks extra work

My advice:

- If you have 4+ weeks: Add Alarm + Economic dataset
- If you have 2-3 weeks: Do the 3 fixes, prepare excellent rebuttal, have backup venue
- Either way: The prompt variation study is absolutely non-negotiable

The variance-aware evaluation framework is genuinely novel - that's your core contribution. The question is whether the execution is comprehensive enough to convince 3 reviewers. With 3 fixes alone, you're in "convincing 2 out of 3" territory. With datasets added, you're in "convincing all 3" territory.

Choose based on your timeline and risk tolerance, but **don't skip the 3 critical fixes under any circumstances**.